BCSE207L // PROGRAMMING FOR DATA SCIENCE

Credit Risk Modeling Using Machine Learning

Explainable loan approval prediction with Random Forest, SHAP and counterfactual recourse

```
model <- randomForest(
  Loan_Status ~ .,
  data = train_data
)

# hold-out test, n = 154
accuracy =    76.62 %
f1       =    83.49 %
auc      =    0.78
```

24BDS1112 · 24BDS1071

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Beyond a label: explainable loan decisions

01 PROBLEM

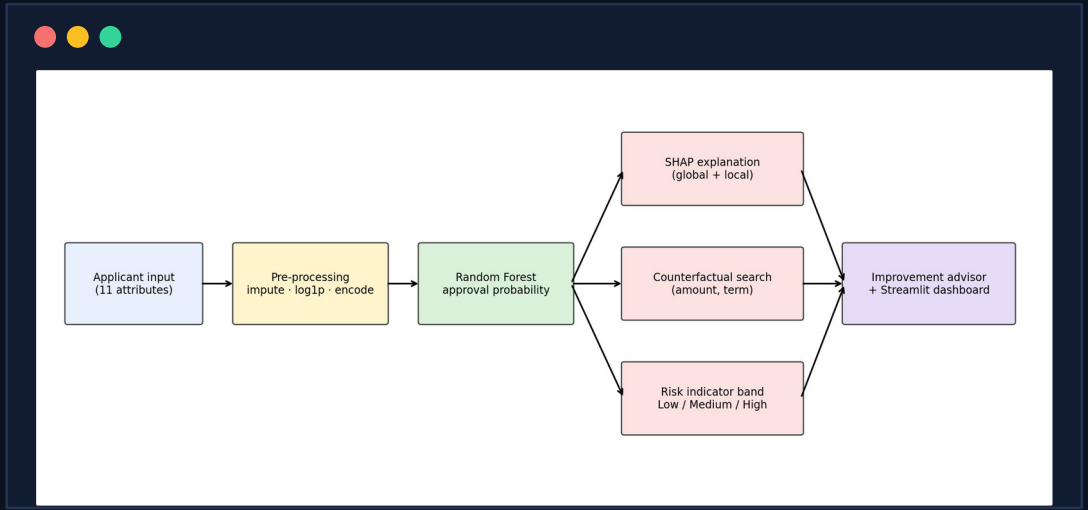
Models return approve/reject with no reason. Applicants can't improve; officers can't audit.

02 GAP

Prior work stops at feature attribution: what mattered, not what to change.

03 APPROACH

Random Forest + SHAP + counterfactuals + advisor + risk band in one Streamlit dashboard.



SHAP
why it decided

COUNTERFACTUAL
what to change

RISK BAND
how likely

// 02 · DATASET & PREPROCESSING

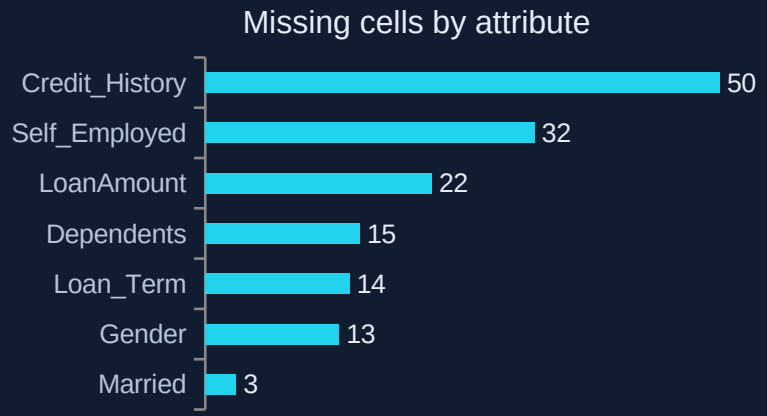
614 applications, imbalanced and messy

614
loan records

11
input features

68.7%
approved (422 / 192)

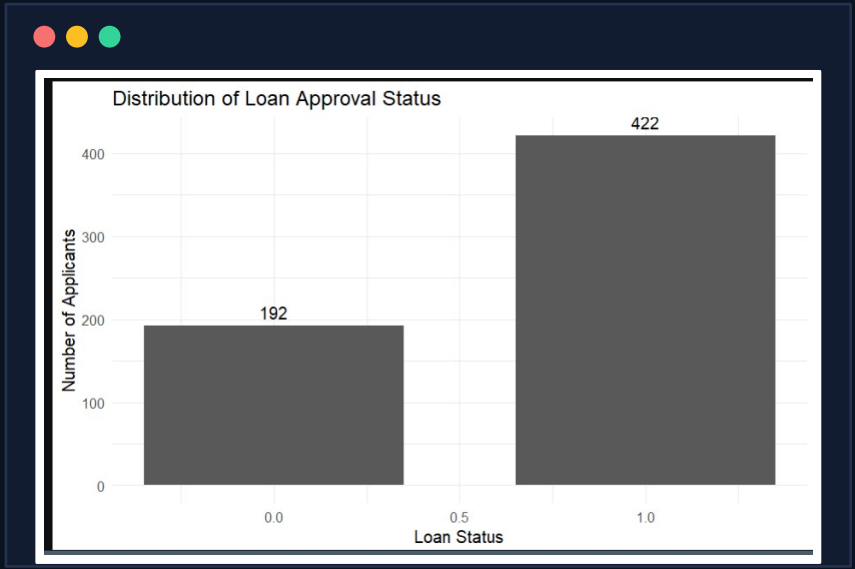
149
missing cells



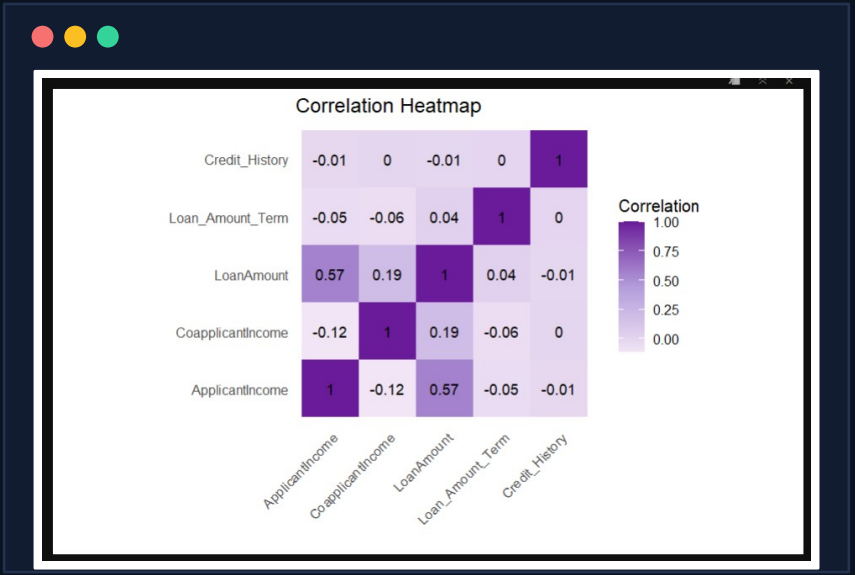
PIPELINE

- Impute: median (numeric), mode (categorical)
- Log transform skewed income (max 81,000)
- Split 75:25 → train 460, test 154
- Oversample train only → 644 rows

EDA: class imbalance and weak correlations



422 approved vs 192 rejected: about 2.2 to 1



Only ApplicantIncome–LoanAmount is notable (0.57)

Four classifiers, one pipeline

LOGISTIC REG.

Baseline

Linear log-odds model. Highly interpretable.

DECISION TREE

Rules

Threshold splits. Easy to read, prone to overfit.

KNN (k = 3)

Distance

Majority vote of 3 nearest applicants. Scale-sensitive.

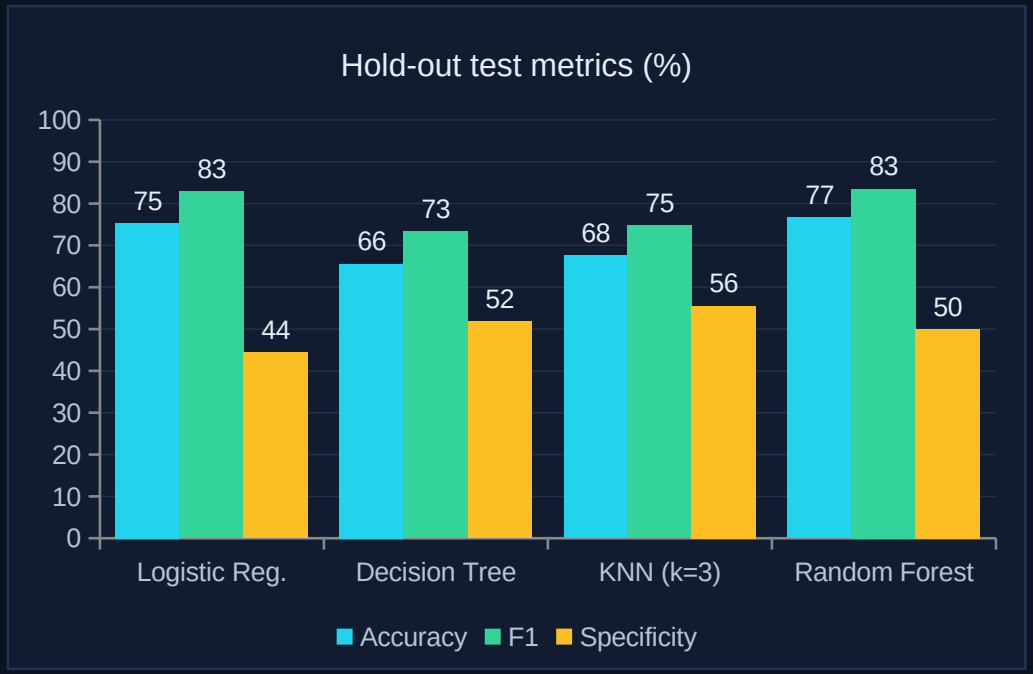
RANDOM FOREST

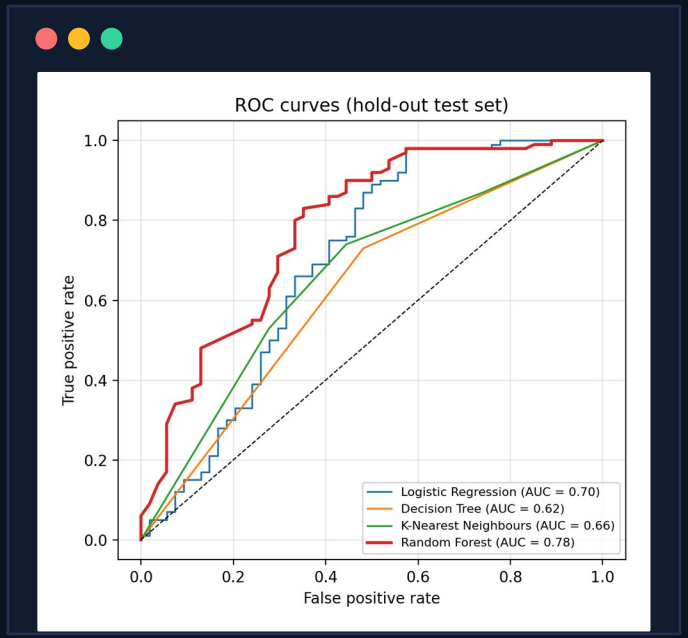
Proposed

Bagged trees, random features. Gives probabilities and exact TreeSHAP.

Default hyperparameters (k = 3 for KNN), fixed seed 42, trained on the oversampled training set.

Random Forest wins on AUC: 0.78





5-fold CV: accuracy 75.89 ± 2.57%, AUC 0.76 ± 0.02. Gap to Logistic Regression is within noise at n = 154.

Hold-out test results (n = 154)

Model	Acc %	Prec %	Rec %	F1 %	Spec %	AUC
Logistic Regression	75.32	75.41	**92.00**	82.88	44.44	0.70
Decision Tree	65.58	73.74	73.00	73.37	51.85	0.62
K-Nearest Neighbours	67.53	75.51	74.00	74.75	**55.56**	0.66
Random Forest (proposed)	**76.62**	**77.12**	91.00	**83.49**	50.00	**0.78**

Green marks the best value per column. Random Forest leads on accuracy, F1 and AUC.

All models favour approvals: high recall, specificity near 50%.

// 07 · CROSS-VALIDATION

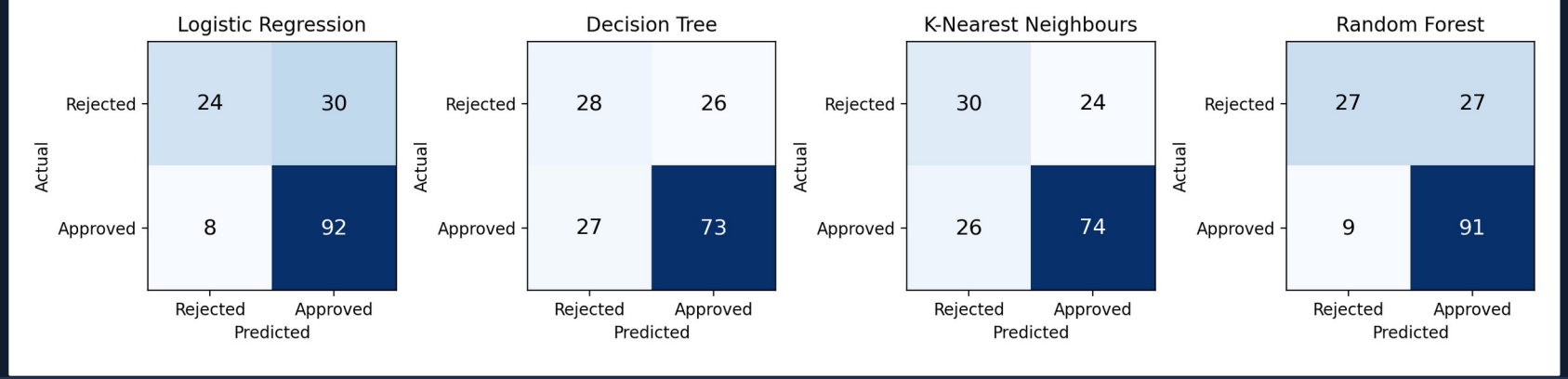
5-fold stratified CV (mean ± std)

Model	Acc %	Prec %	Rec %	F1 %	Spec %	AUC
Logistic Regression	75.58 ± 4.37	79.27 ± 0.96	87.21 ± 7.04	82.97 ± 3.67	49.97 ± 2.75	0.71 ± 0.05
Decision Tree	71.16 ± 3.30	79.11 ± 1.36	78.89 ± 6.09	78.90 ± 3.21	**54.16 ± 5.09**	0.67 ± 0.02
K-Nearest Neighbours	69.05 ± 2.91	78.49 ± 2.00	75.81 ± 5.04	77.04 ± 2.78	54.09 ± 7.37	0.69 ± 0.02
Random Forest (proposed)	**75.89 ± 2.57**	**79.51 ± 0.74**	**87.42 ± 4.14**	**83.25 ± 2.19**	50.51 ± 1.99	**0.76 ± 0.02**

RF and Logistic Regression are indistinguishable on accuracy; RF has the higher, steadier AUC.

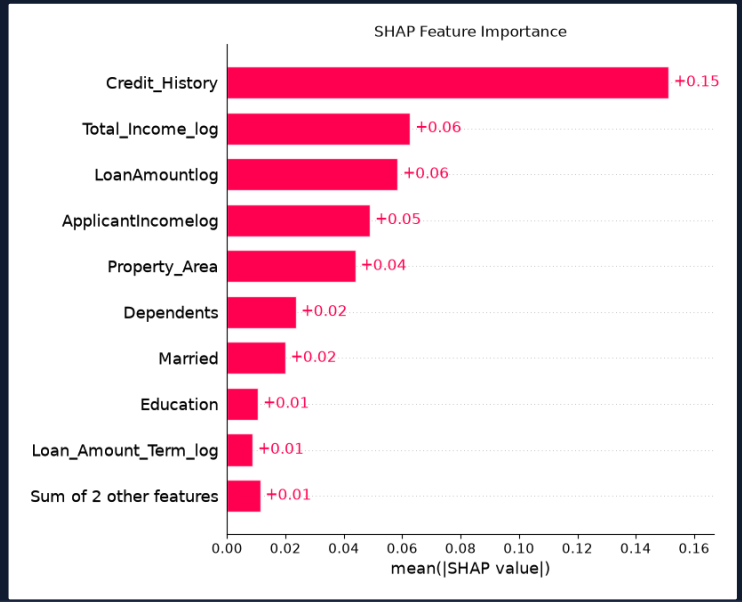
Oversampling was applied inside each training fold only.

Where each model errs

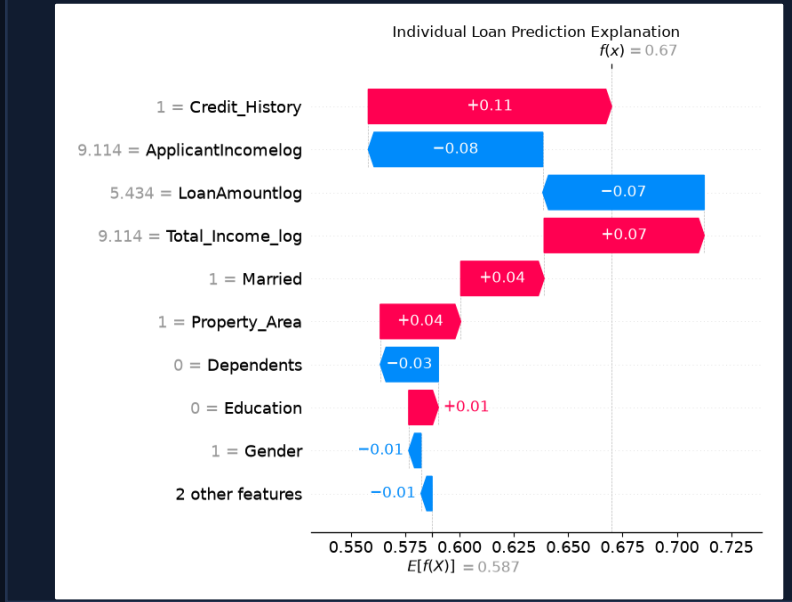


Logistic Reg.	Decision Tree	KNN (k = 3)	Random Forest
FP 30 · FN 8	**FP 26 · FN 27**	**FP 24 · FN 26**	**FP 27 · FN 9**

SHAP: credit history dominates decisions

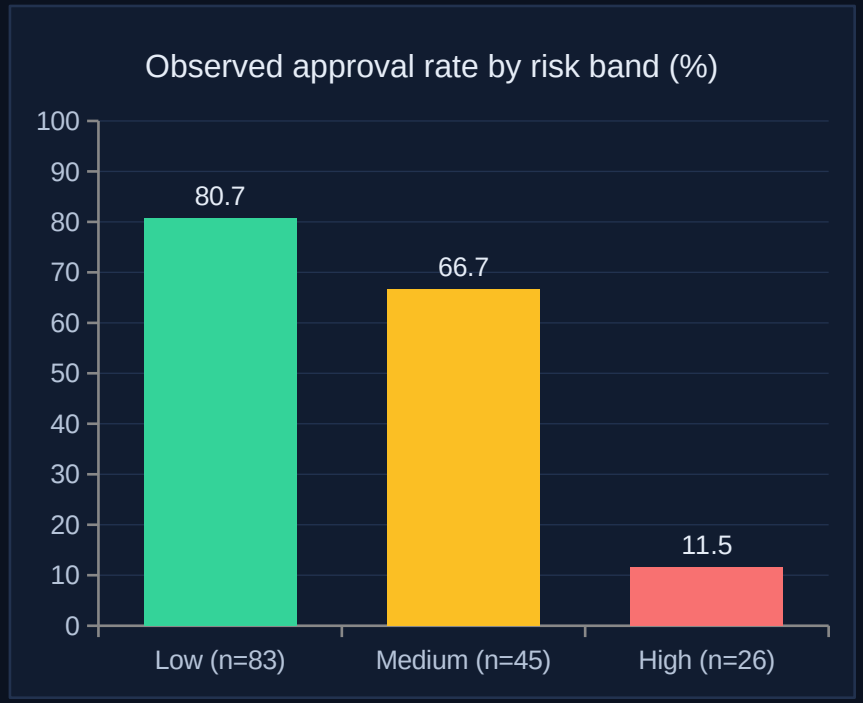


Global: Credit_History 0.15, about 2.5× the next feature



Local: one applicant, p = 0.67 (Medium band)

From explanation to recourse



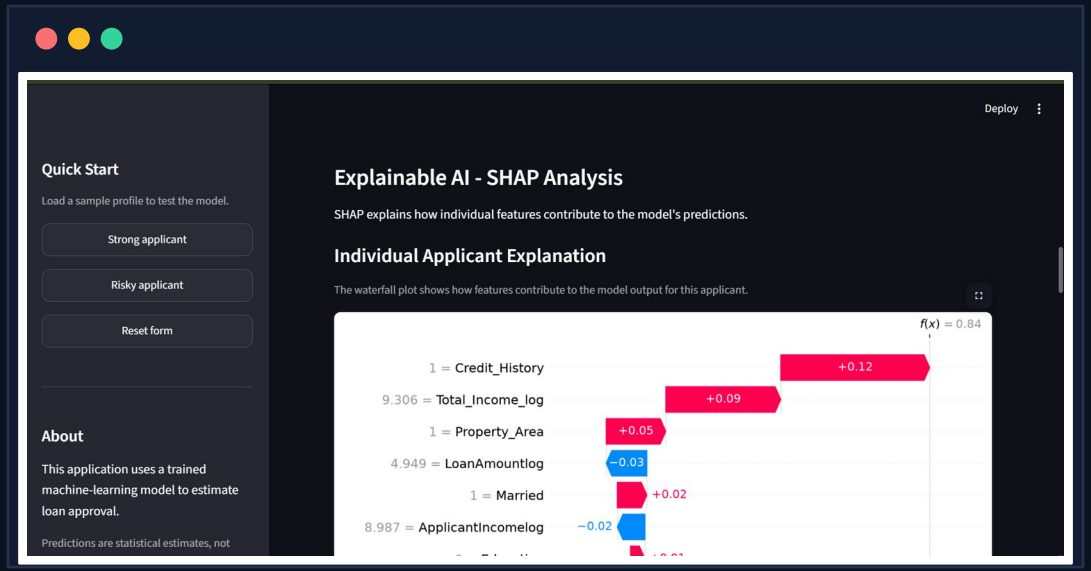
6 / 35 rejected applicants flipped to approved (17.1%)

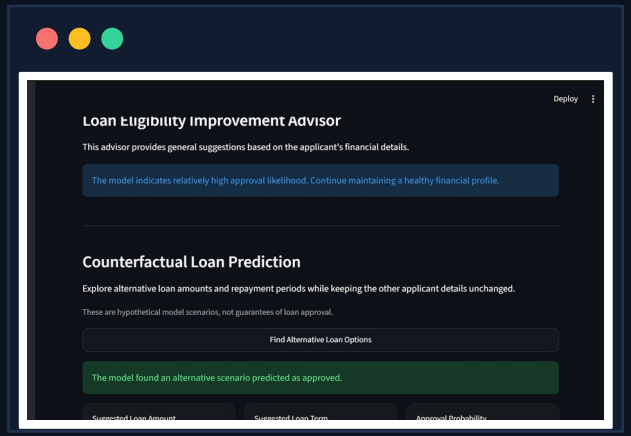
−23.1% mean loan reduction needed (range 9.3–49.7%)

0.447→0.517 mean approval probability, before → after

23 of 35 rejections have poor credit history; no loan change offsets it.

Streamlit app for non-technical users





Inputs → decision, probability, risk band, EMI summary, SHAP plots, counterfactuals and plain-language advice.

Limits and what comes next

LIMITATIONS

- Small data: 154 test rows, CV std 2–4 points
- Specificity ≈ 50%: not for unsupervised decisions
- Gender and marital status are inputs; historical bias can carry over
- Bands show approval likelihood, not default risk

TAKEAWAYS & NEXT

- The contribution is the integrated framework, not an accuracy jump
- Validate on larger multi-lender data; audit fairness
- Widen counterfactuals to credit history and co-applicant income
- Test gradient boosting with the same explainers